TARUN SINGH

PhD Scholar | Department of Civil Engineering | Indian Institute of Technology Roorkee, India
Academic webpage | Google Scholar | ORCID

RESEARCH PROFILE

PhD researcher in constitutive modelling of rate-dependent materials, with emphasis on finite-deformation continuum mechanics, rheological connections, evolving natural configurations, and thermodynamically consistent modelling of viscoelastic solids. Research combines constitutive theory, numerical integration, parameter identification, and computational implementation.

RESEARCH INTERESTS

Finite-deformation continuum mechanics and constitutive theory • Finite viscoelasticity and viscoplasticity • Evolving natural configurations and rheological connections • Geometric perspectives in continuum mechanics • Anisotropic constitutive modelling of rate-dependent materials • Thermodynamics and numerical implementation of constitutive models

EDUCATION

PhD in Civil Engineering | 2023–Present
Indian Institute of Technology Roorkee, India
Research: Constitutive modelling of rate-dependent materials | Supervisor: Dr. Sandipan Paul

M.Tech. in Structural and construction Engineering | 2020–2022
Dr. B. R. Ambedkar National Institute of Technology Jalandhar, India
Thesis: Carbonation resistance of concrete made with iron slag and recycled concrete aggregate

B.Tech. in Civil Engineering | 2015–2019
Rewa Engineering College, Rewa, Madhya Pradesh, India

RESEARCH EXPERIENCE AND CONTRIBUTIONS

- Developed finite-deformation constitutive frameworks for rate-dependent materials, with a focus on viscoelastic solids.
- Extended rheological connections to finite deformations using multiple natural configurations.
- Formulated Kelvin–Voigt-type viscoelastic constitutive models using the theory of evolving natural configurations.
- Developed a deformation-mode-specific constitutive framework based on scalar, conjugate stress/strain pairs and applied it to soft-material characterization.
- Performed numerical integration of nonlinear constitutive equations and parameter identification using experimental response data.

JOURNAL PUBLICATIONS – DOCTORAL RESEARCH

1. **T. Singh** and S. Paul (2026). *"A deformation mode-specific constitutive framework for finite viscoelasticity based on scalar, conjugate stress/strain pairs: Applications to characterization of soft materials."* Journal of the Mechanics and Physics of Solids, Article 106895. DOI
2. **T. Singh** and S. Paul (2026). *"Constitutive modeling of Kelvin–Voigt type viscoelastic solids undergoing finite deformations using the theory of evolving natural configurations."* International Journal of Solids and Structures, 341, Article 114336. DOI
3. **T. Singh** and S. Paul (2025). *"On the extension of the concept of rheological connections to a finite deformation framework using multiple natural configurations."* International Journal of Engineering Science, 212, Article 104267. DOI

EARLIER PUBLICATIONS – M.TECH. RESEARCH

1. N. Singh, A. Singh, N. Ankur, P. Kumar, M. Kumar, and **T. Singh** (2022). *"Reviewing the properties of recycled concrete aggregates and iron slag in concrete."* Journal of Building Engineering, 60, 105150.
2. N. Singh, N. Ankur, **T. Singh**, M. Kumar, A. Singh, and P. Kumar (2022). *"Non-destructive evaluation of iron slag concrete containing recycled concrete aggregates."* Indian Concrete Journal, 96(11).
3. N. Singh, **T. Singh**, M. Kumar, A. Singh, and P. Kumar (2022). *"Investigating the fresh state performance of concrete containing iron slag and recycled concrete aggregates."* Materials Today: Proceedings, 65, 1467–1477.

CONFERENCE PRESENTATIONS

- T. Singh and S. Paul (2026). "Constitutive modeling of viscoelastic solids undergoing finite deformations using the theory of evolving natural configurations." IUTAM Symposium: Rate Dependent Mechanics and Multiscale Modelling of Materials, IIT Delhi.
- T. Singh and S. Paul (2026). "Thermodynamically consistent constitutive model for viscoelastic solids under finite deformation using evolving natural configurations." 7th Indian Conference on Applied Mechanics, IIT Kanpur.
- T. Singh and S. Paul (2025). "Challenges in extending rheological connections to a finite deformation theory." 14th International Symposium on Plasticity and Impact Mechanics (IMPLAST 2025), IIT Roorkee.

COMPUTATIONAL AND TECHNICAL SKILLS

Programming: MATLAB, Fortran

Finite-element analysis: Abaqus/Standard

Constitutive implementation: Finite-strain viscoelastic UMAT development and verification

Numerical methods: Numerical integration of constitutive equations; parameter identification and model calibration

ADDITIONAL RESEARCH DIRECTIONS

- Implemented finite-strain viscoelastic constitutive response in an Abaqus/Standard UMAT using Fortran.
- Current research directions include extensions towards viscoplasticity and anisotropic formulations for other material symmetry classes.